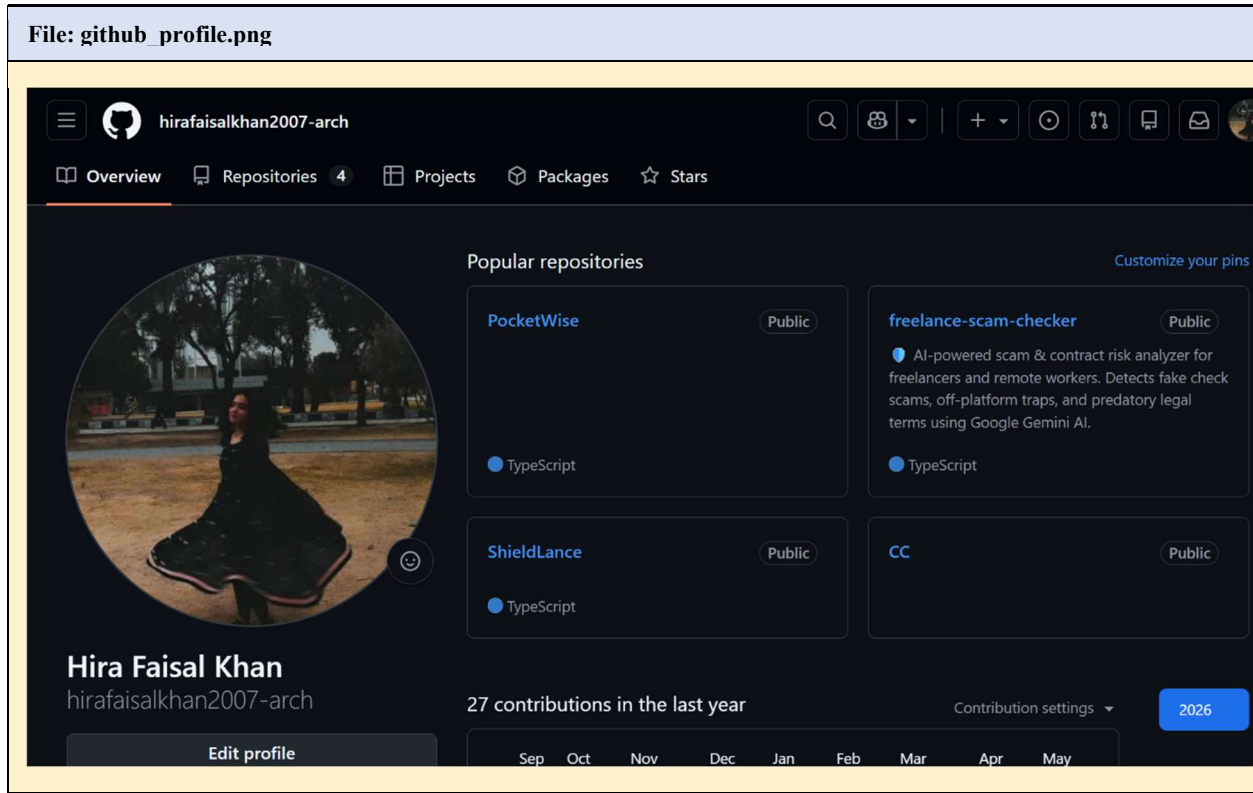


Lab 01

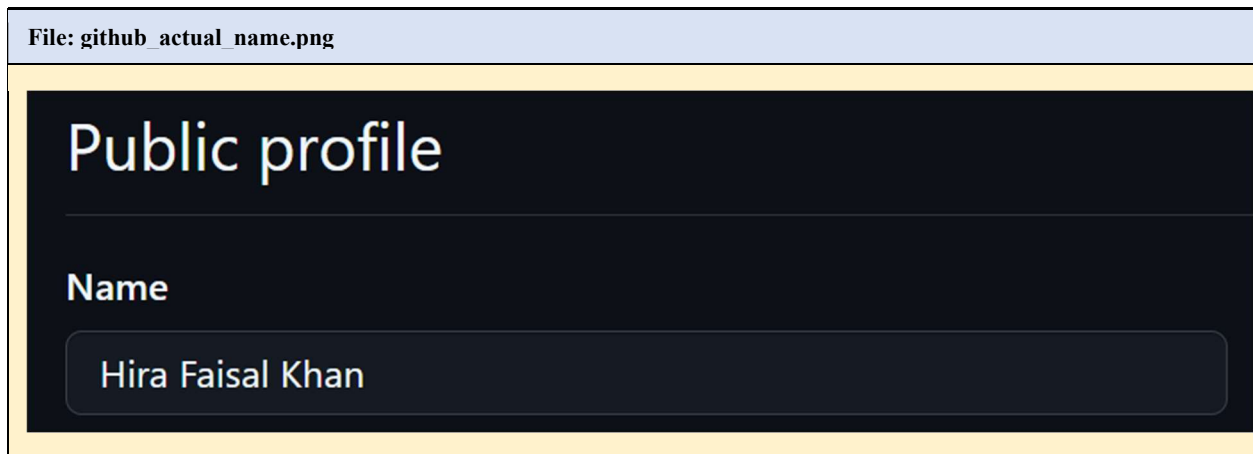
GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name	Hira Faisal Khan
Registration Number	2024-BSE-07
Course Code	SE-665
Section	A
GitHub Username	hirafaisalkhan2007-arch

Task 1: Create and Configure Your GitHub Account



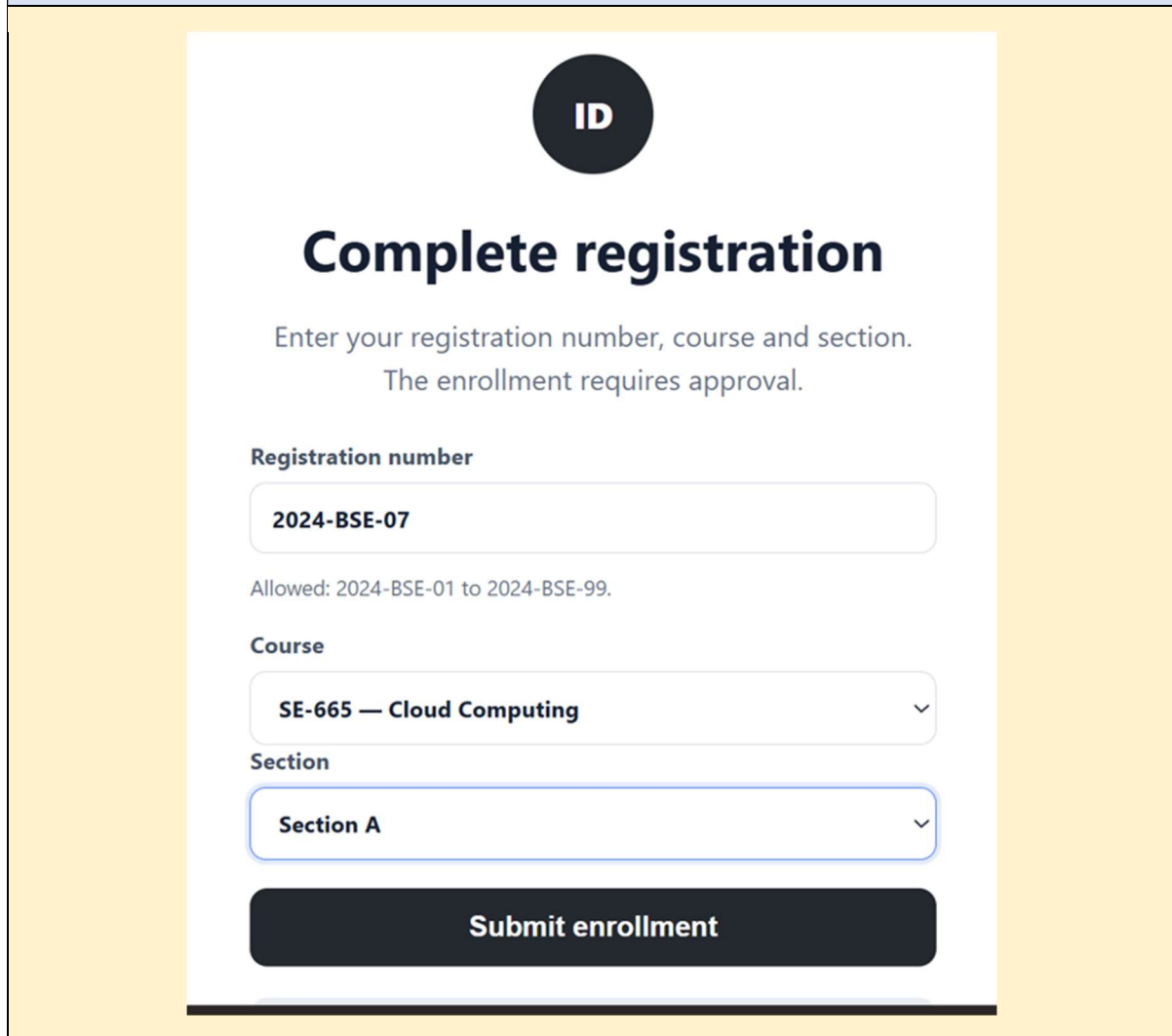
Caption: This screenshot shows my GitHub profile page with my actual name and GitHub username visible.



Caption: This screenshot shows the Name field under GitHub Settings > Public profile filled in with my complete actual name.

Task 2: Register in the Student Marks Portal

File: portal_login_page.png



The screenshot shows a registration page with a yellow background. At the top center is a dark circle with the text 'ID'. Below it is the heading 'Complete registration' in bold. Under the heading is the text 'Enter your registration number, course and section. The enrollment requires approval.' There are three input fields: 'Registration number' with the value '2024-BSE-07', 'Course' with the value 'SE-665 — Cloud Computing', and 'Section' with the value 'Section A'. Below these fields is a dark button labeled 'Submit enrollment'.

ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-07

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

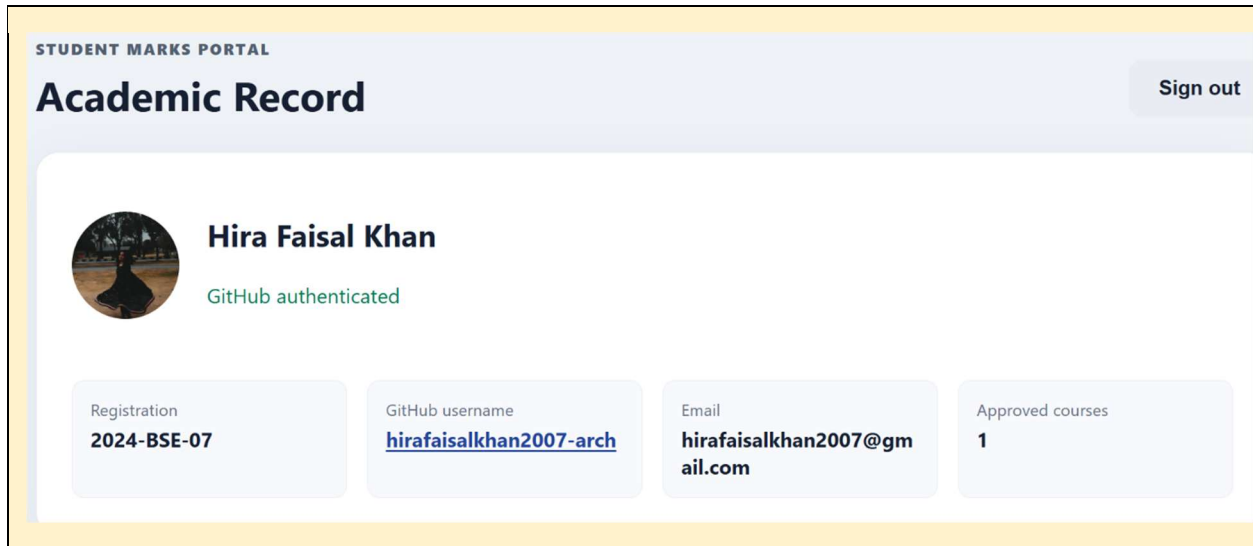
Section

Section A

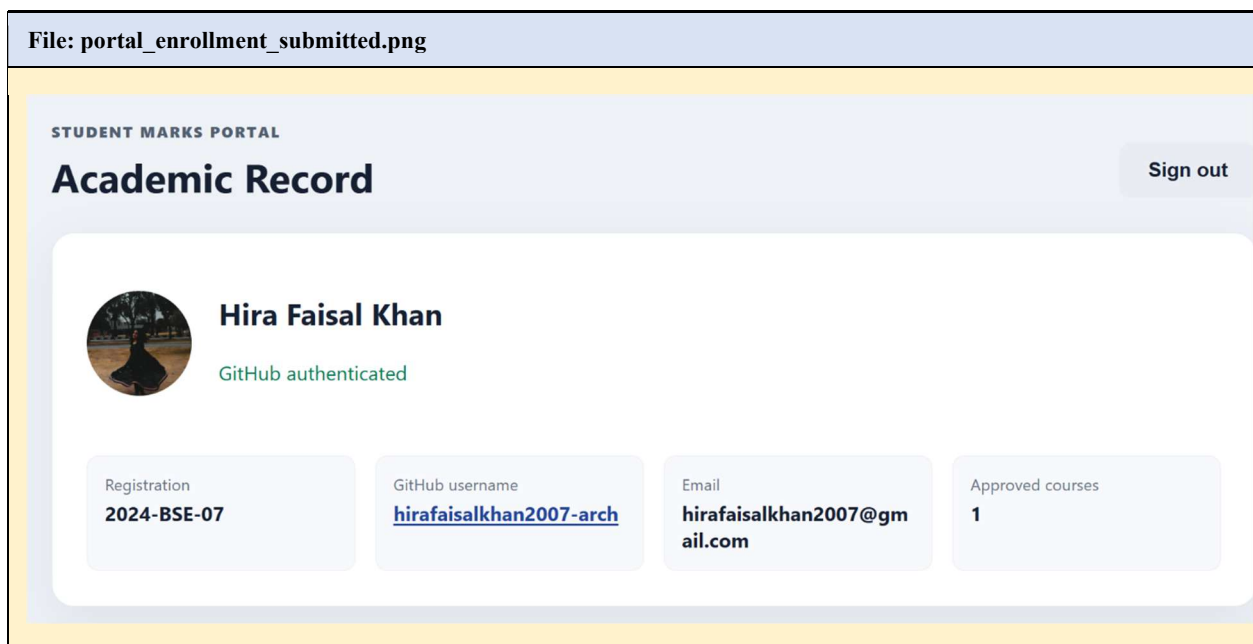
Submit enrollment

Caption: This screenshot shows the Student Marks Portal login page before signing in.

File: portal_github_profile.png

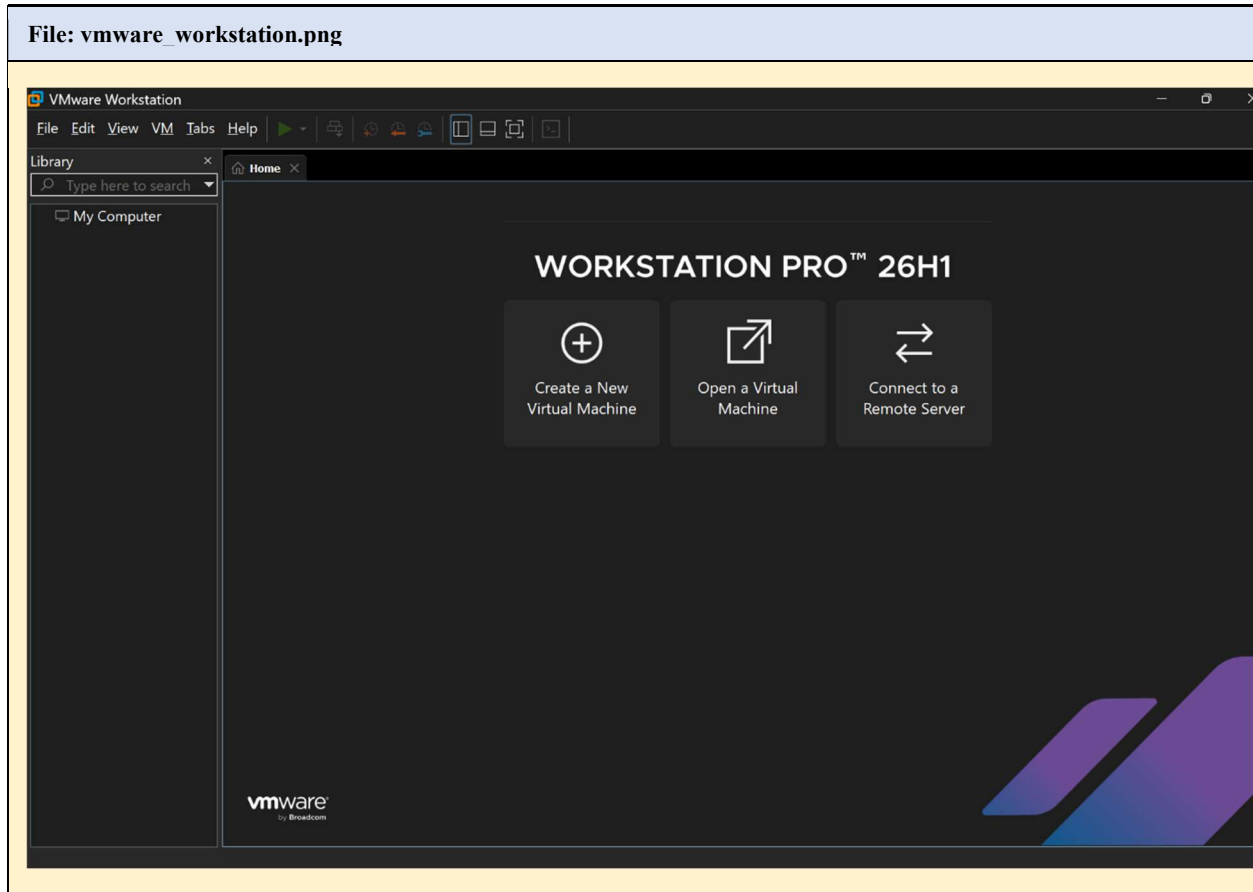


Caption: This screenshot shows my GitHub profile information as it appears inside the Student Marks Portal after signing in.

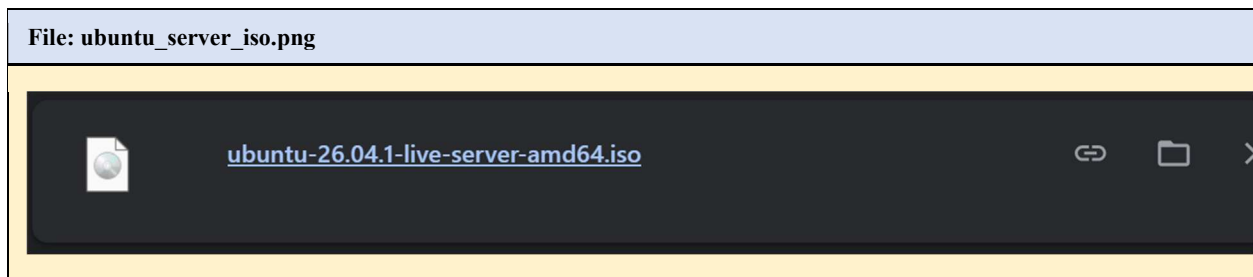


Caption: This screenshot shows my enrollment submitted in the portal with my registration number, course code, and section.

Task 3: Download VMware Workstation Pro and Ubuntu Server

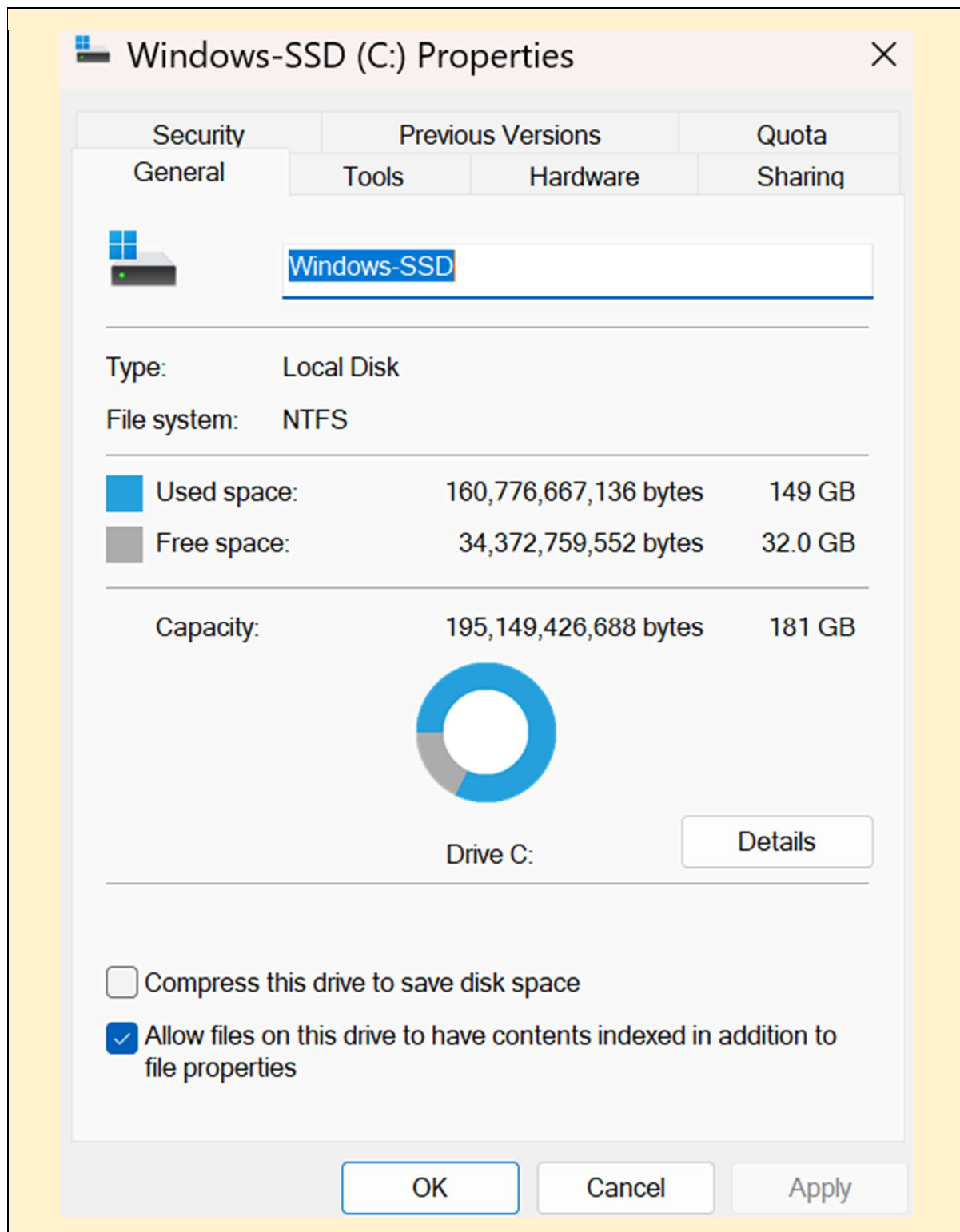


Caption: This screenshot shows VMware Workstation Pro open and running on my computer.



Caption: This screenshot shows the downloaded Ubuntu Server LTS ISO file and its filename.

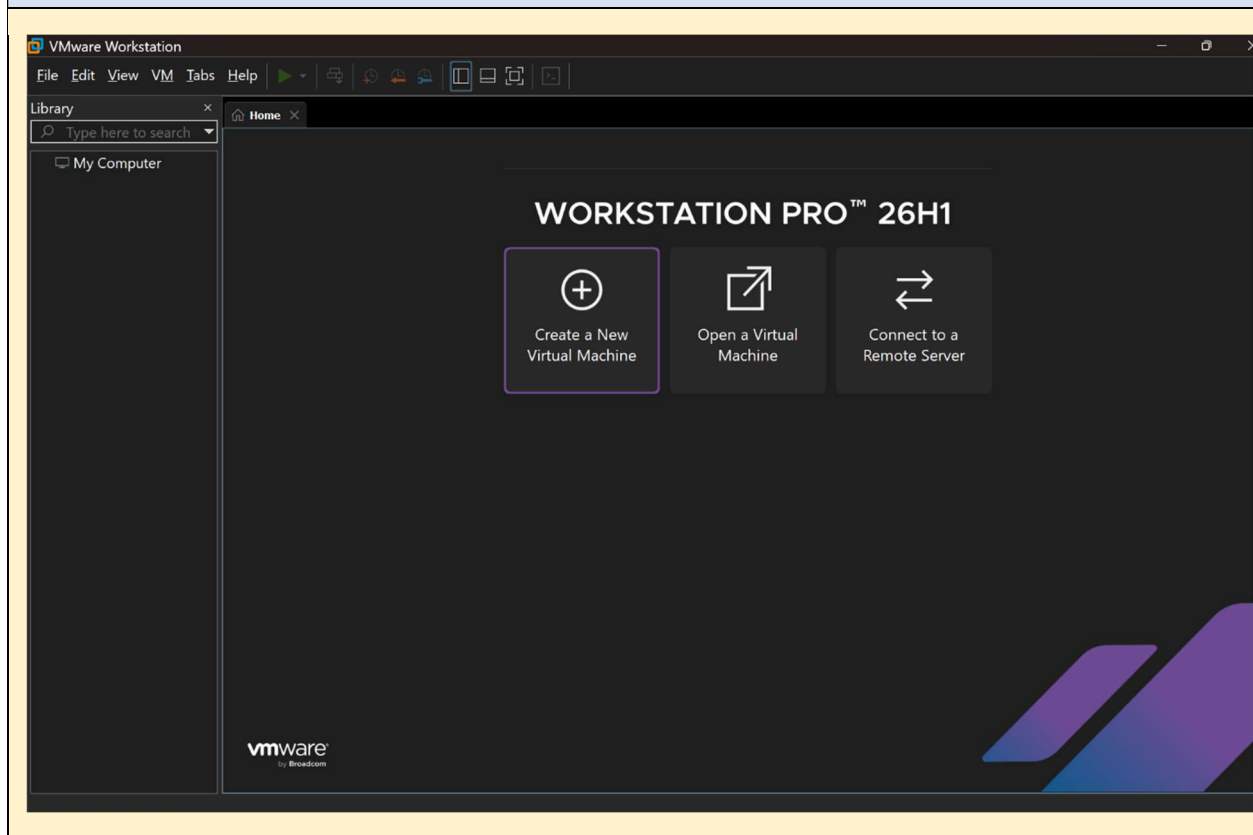




Caption: This screenshot shows at least 20 GB of free storage available for the Ubuntu Server virtual machine.

Task 4: Create the Ubuntu Server Virtual Machine

File: vm_creation_wizard.png



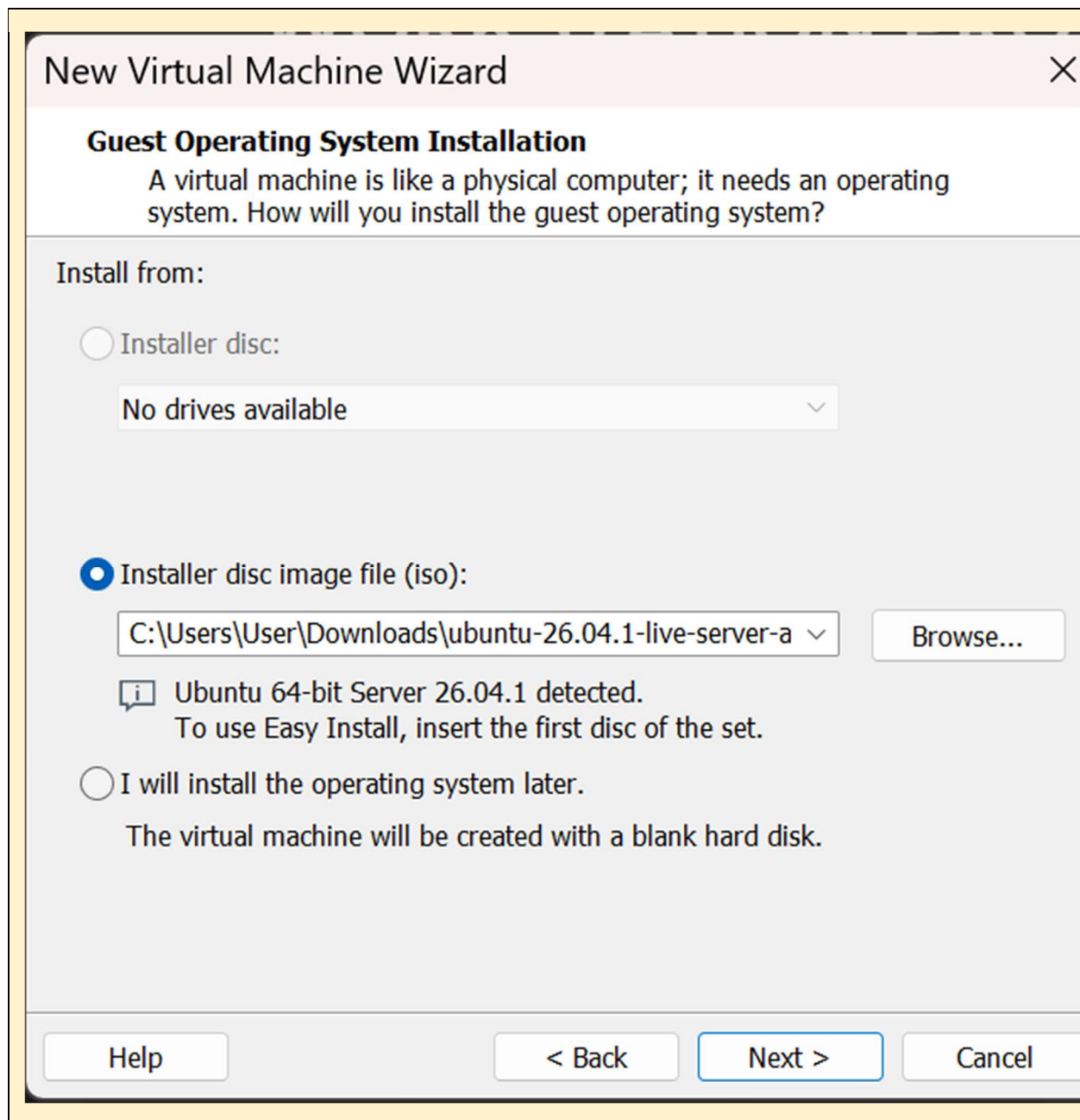
Caption: This screenshot shows the New Virtual Machine Wizard opened in VMware Workstation Pro.

File: vm_typical_configuration.png



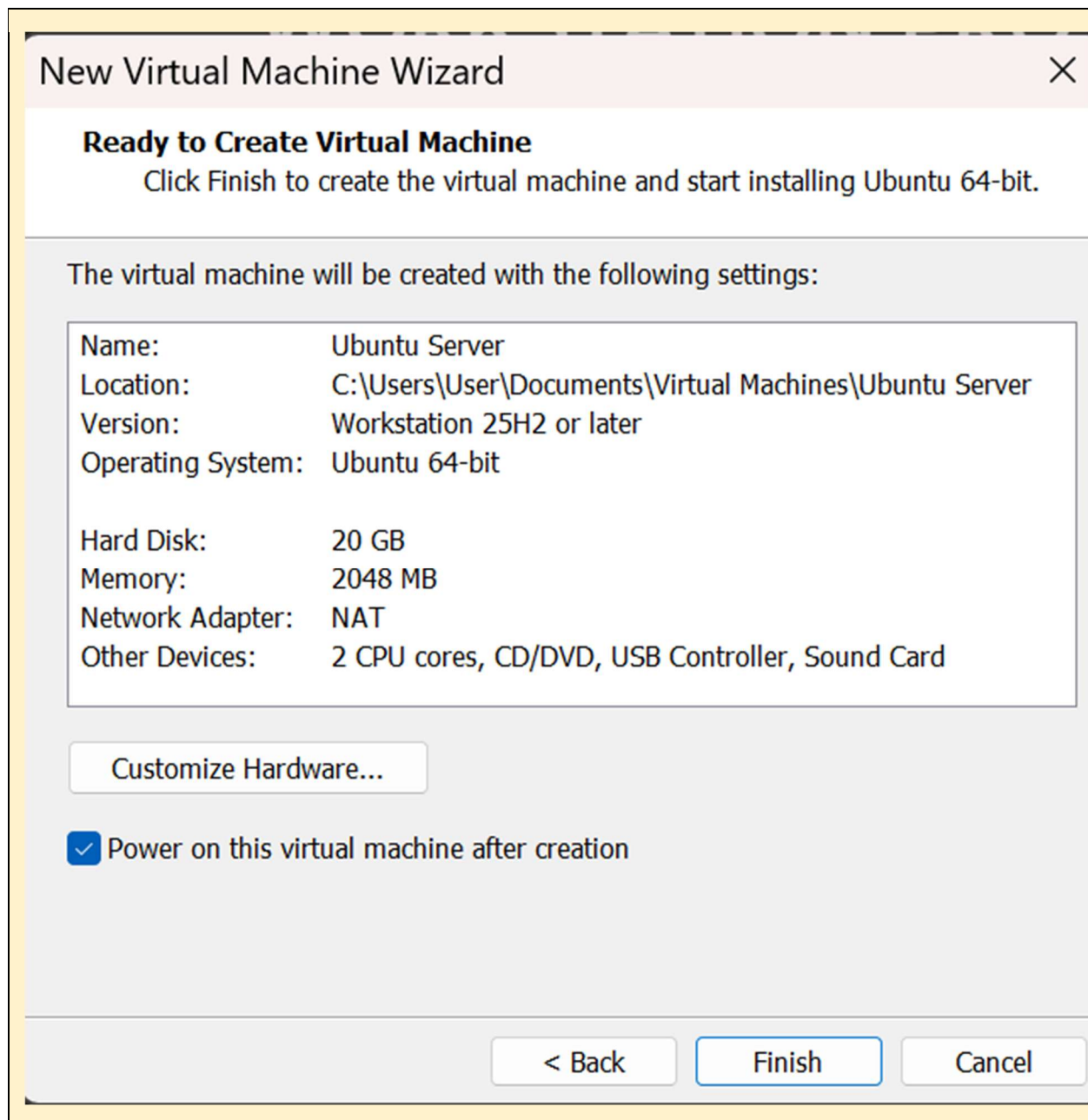
Caption: This screenshot shows the 'Typical (recommended)' virtual machine configuration option selected.

File: vm_ubuntu_iso_selected.png



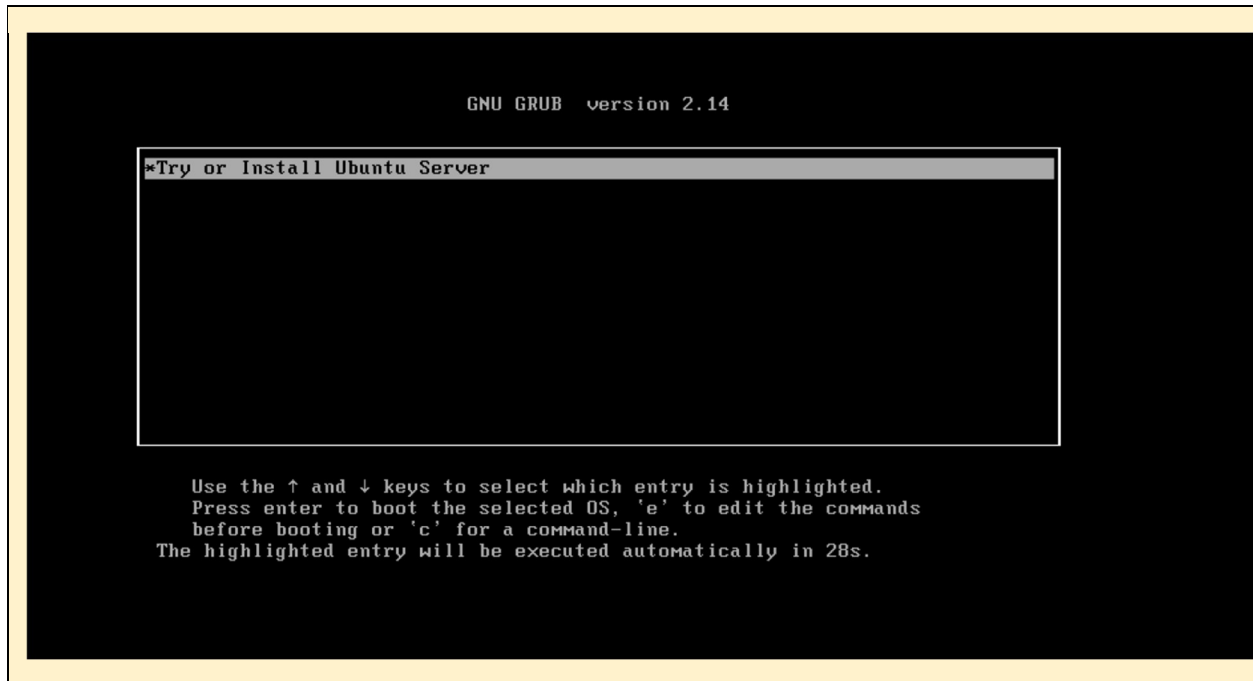
Caption: This screenshot shows the Ubuntu Server ISO file selected as the installer disc image.

File: vm_configuration_summary.png



Caption: This screenshot shows the final virtual machine configuration summary with at least 20 GB of allocated disk space.

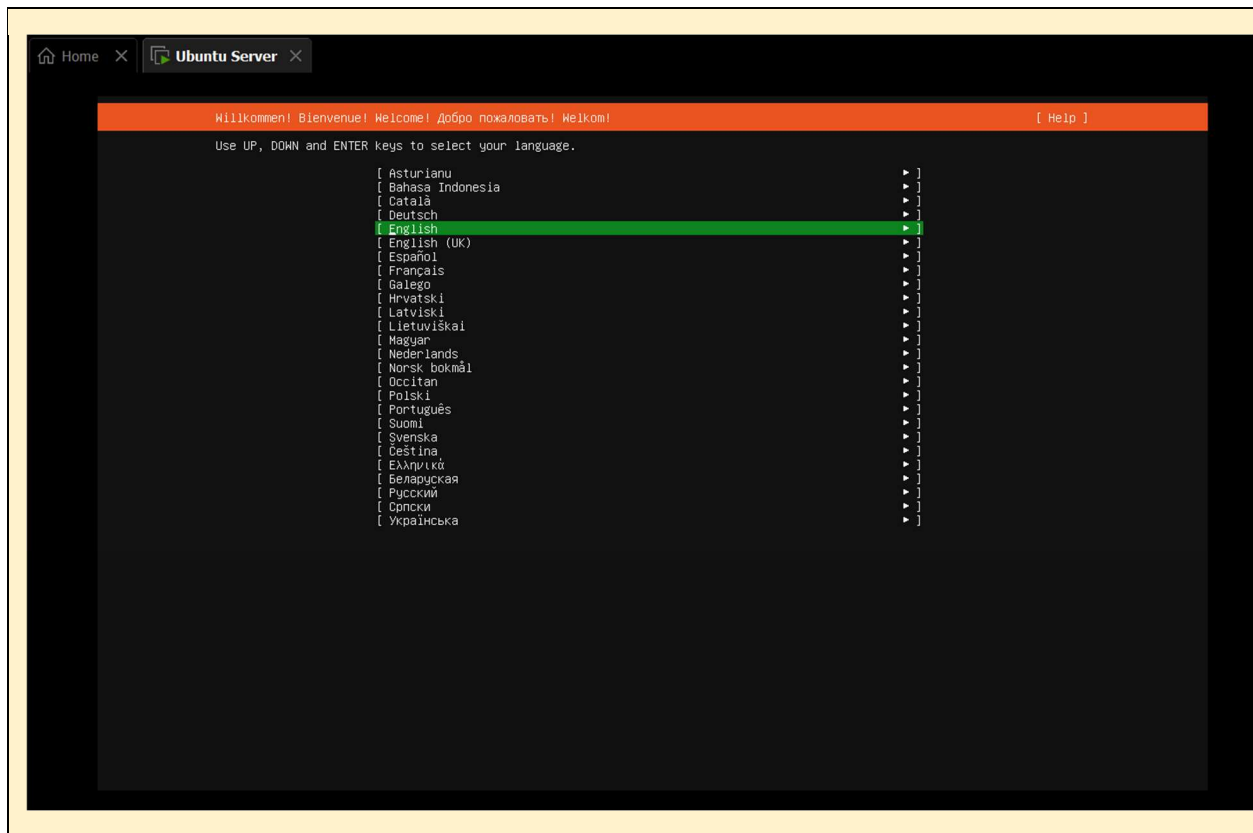
File: ubuntu_installer_boot.png



Caption: This screenshot shows the Ubuntu Server installer boot screen after powering on the virtual machine.

Task 5: Choose the Installation Language

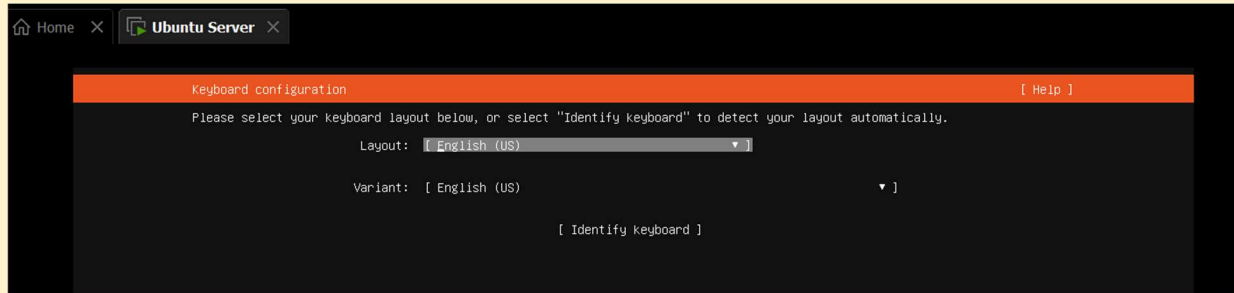
File: ubuntu_language.png



Caption: This screenshot shows the installation language selected in the Ubuntu Server installer.

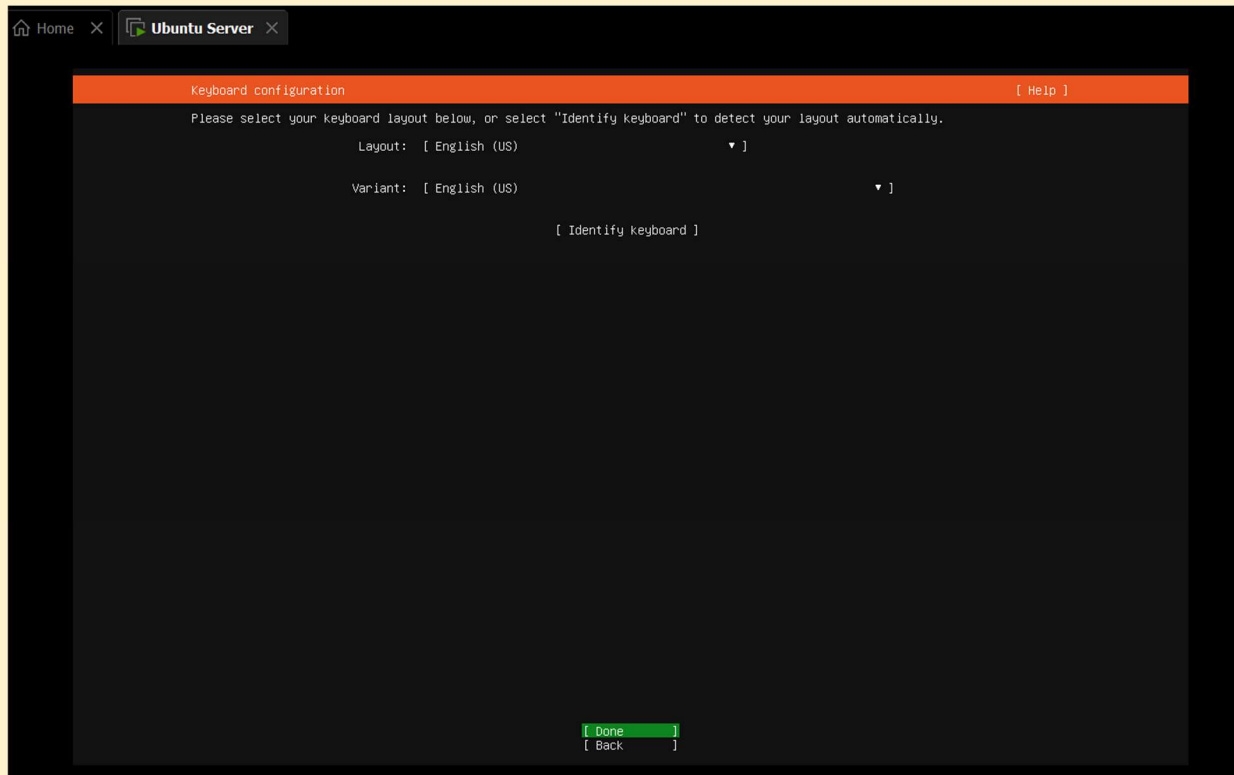
Task 6: Configure the Keyboard Layout

File: ubuntu_keyboard_layout.png



Caption: This screenshot shows the keyboard layout selected in the installer.

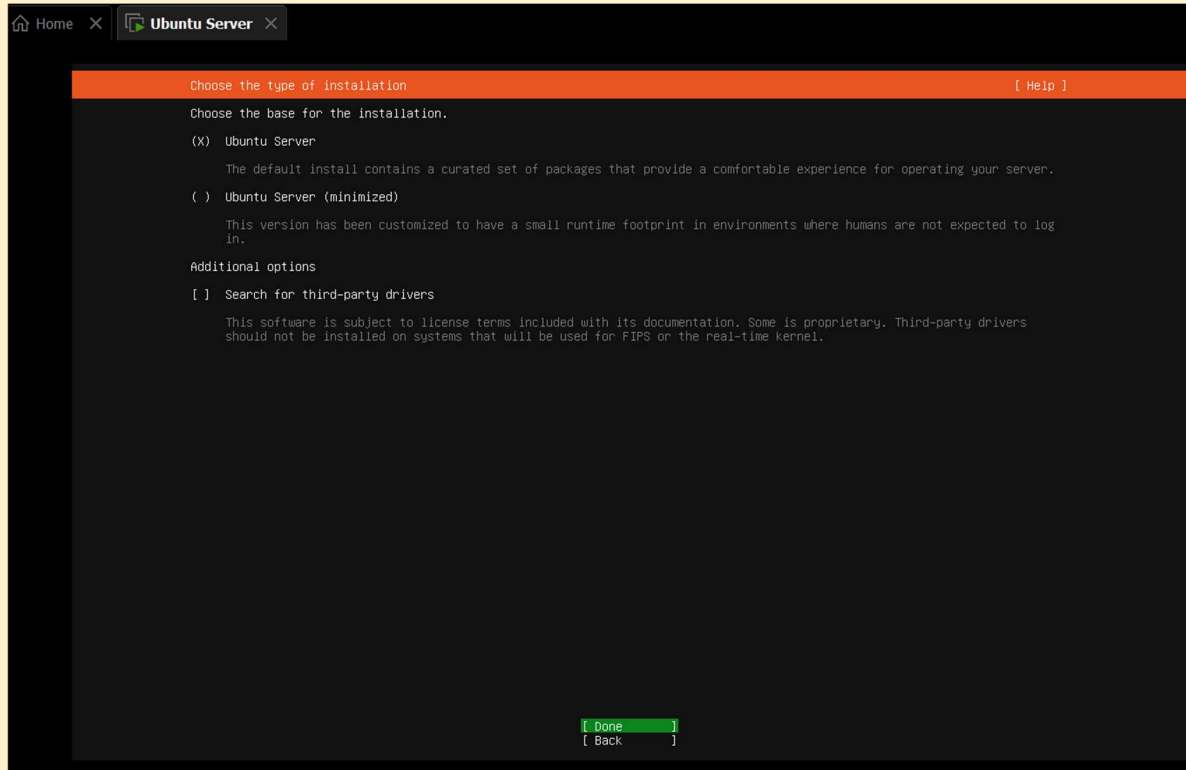
File: ubuntu_keyboard_variant.png



Caption: This screenshot shows the keyboard variant selected in the installer.

Task 7: Choose the Ubuntu Server Installation Type

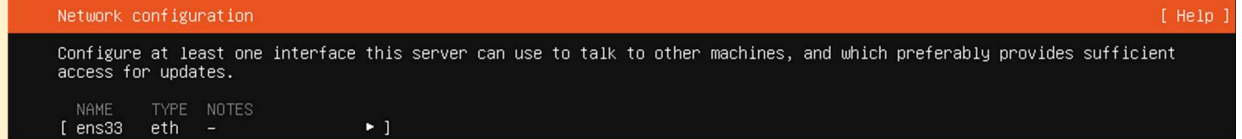
File: ubuntu_installation_type.png



Caption: This screenshot shows the normal Ubuntu Server installation option selected, not MAAS.

Task 8: Configure the Network

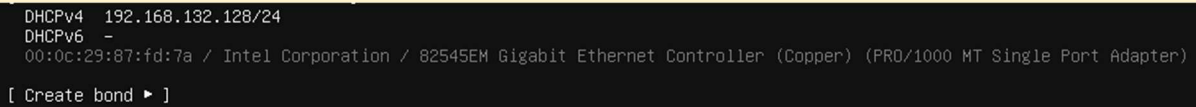
File: ubuntu_network_interface.png



NAME	TYPE	NOTES
ens33	eth	-

Caption: This screenshot shows the network interface detected by the Ubuntu Server installer.

File: ubuntu_installer_network.png



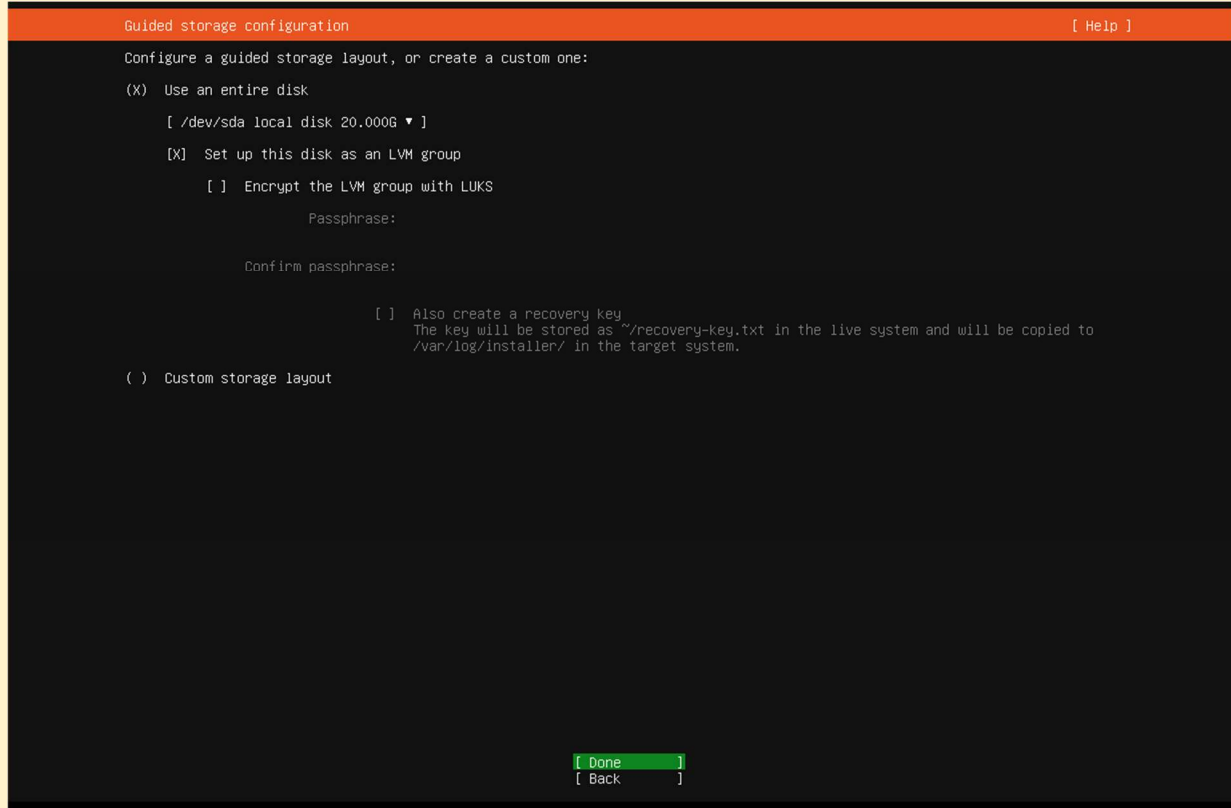
[Create bond ►]

[Done]
[Back]

Caption: This screenshot shows the network interface has received an IP address automatically through DHCP.

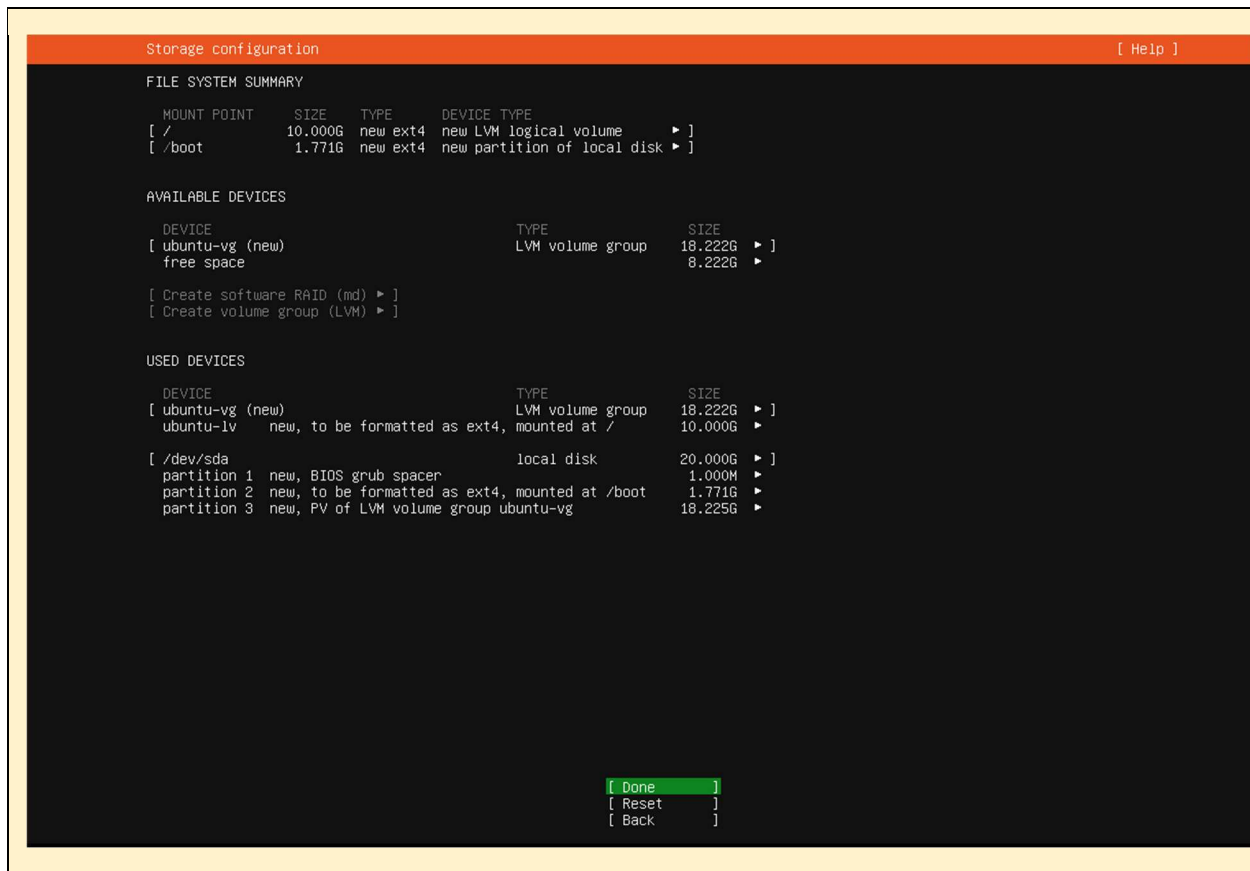
Task 9: Configure Storage

File: ubuntu_guided_storage.png



Caption: This screenshot shows the guided storage option selected, using the VMware virtual disk.

File: ubuntu_storage_configuration.png



Caption: This screenshot shows the storage configuration reviewed before continuing with installation.

Task 10: Select the Installation Disk

File: ubuntu_selected_disk.png

```
[ /dev/sda                                local disk                20.000G ▶  
partition 1 new, BIOS grub spacer          1.000M ▶  
partition 2 new, to be formatted as ext4, mounted at /boot 1.771G ▶  
partition 3 new, PV of LVM volume group ubuntu-vg 18.225G ▶
```

Caption: This screenshot shows the virtual disk selected for the Ubuntu Server virtual machine.

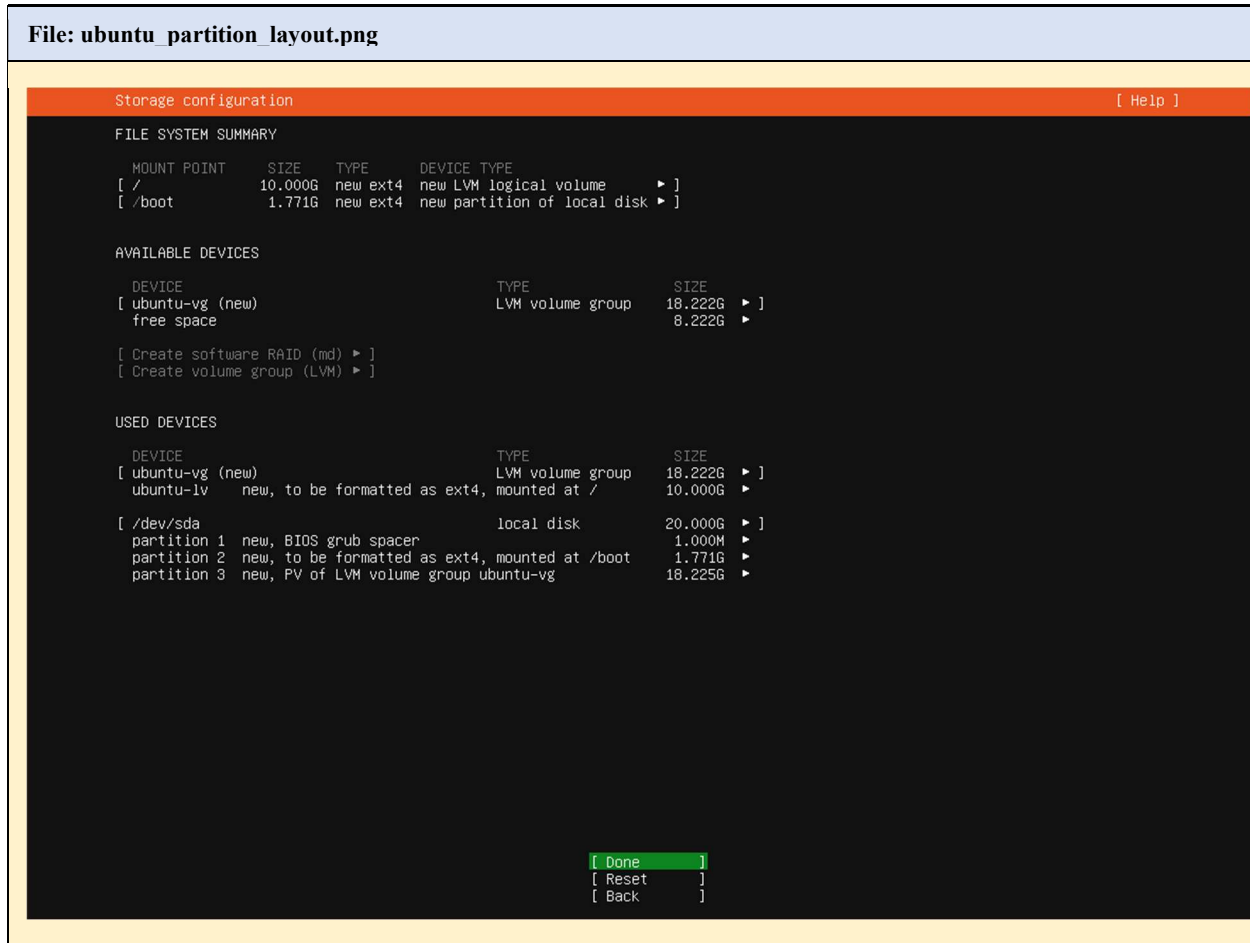
File: ubuntu_disk_details.png

```
Storage configuration [ Help ]  
  
FILE SYSTEM SUMMARY  
  
MOUNT POINT  SIZE  TYPE  DEVICE TYPE  
[ /           10.000G new ext4 new LVM logical volume ▶ ]  
[ /boot       1.771G new ext4 new partition of local disk ▶ ]  
  
AVAILABLE DEVICES  
  
DEVICE              TYPE              SIZE  
[ ubuntu-vg (new)    LVM volume group  18.222G ▶ ]  
free space          8.222G ▶  
  
[ Create software RAID (md) ▶ ]  
[ Create volume group (LVM) ▶ ]  
  
USED DEVICES  
  
DEVICE              TYPE              SIZE  
[ ubuntu-vg (new)    LVM volume group  18.222G ▶ ]  
ubuntu-lv           new, to be formatted as ext4, mounted at / 10.000G ▶  
  
[ /dev/sda          local disk        20.000G ▶ ]  
partition 1 new, BIOS grub spacer          1.000M ▶  
partition 2 new, to be formatted as ext4, mounted at /boot 1.771G ▶  
partition 3 new, PV of LVM volume group ubuntu-vg 18.225G ▶  
  
[ Done ]  
[ Reset ]  
[ Back ]
```

Caption: This screenshot shows the disk name and size, confirming it is the correct VMware virtual disk.

Task 11: Review the Partitions

File: ubuntu_partition_layout.png



Caption: This screenshot shows the proposed partition layout belonging to the VMware virtual disk.

Task 12: Confirm the Storage Changes

File: ubuntu_storage_warning.png

```
e RAID (md) ► ]
group (LVM) ► ]

          Confirm destructive action

Selecting Continue below will begin the installation process and
result in the loss of data on the disks selected to be formatted.

You will not be able to return to this or a previous screen once the
installation has started.

Are you sure you want to continue?

[ No ]
[ Continue ]
```

Caption: This screenshot shows the final destructive-action warning displayed before confirming storage changes.

File: ubuntu_storage_confirmed.png

```
ID (md) ► ]
o (LVM) ► ]

          Confirm destructive action

Selecting Continue below will begin the installation process and
result in the loss of data on the disks selected to be formatted.

You will not be able to return to this or a previous screen once the
installation has started.

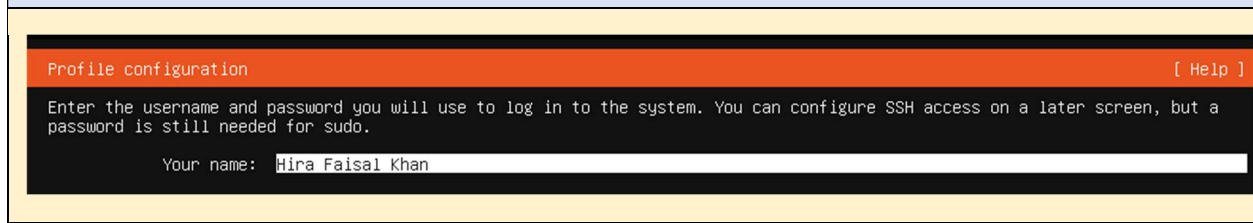
Are you sure you want to continue?

[ No ]
[ Continue ]
```

Caption: This screenshot shows the storage changes confirmed after verifying the correct disk was selected.

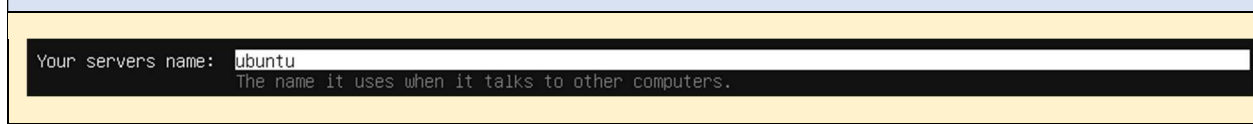
Task 13: Configure the Ubuntu Profile

File: ubuntu_actual_name.png



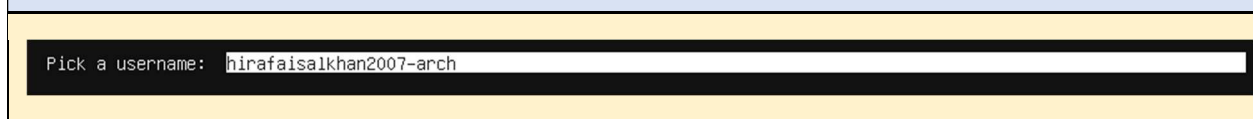
Caption: This screenshot shows my complete actual name entered in the 'Your name' field.

File: ubuntu_server_name.png



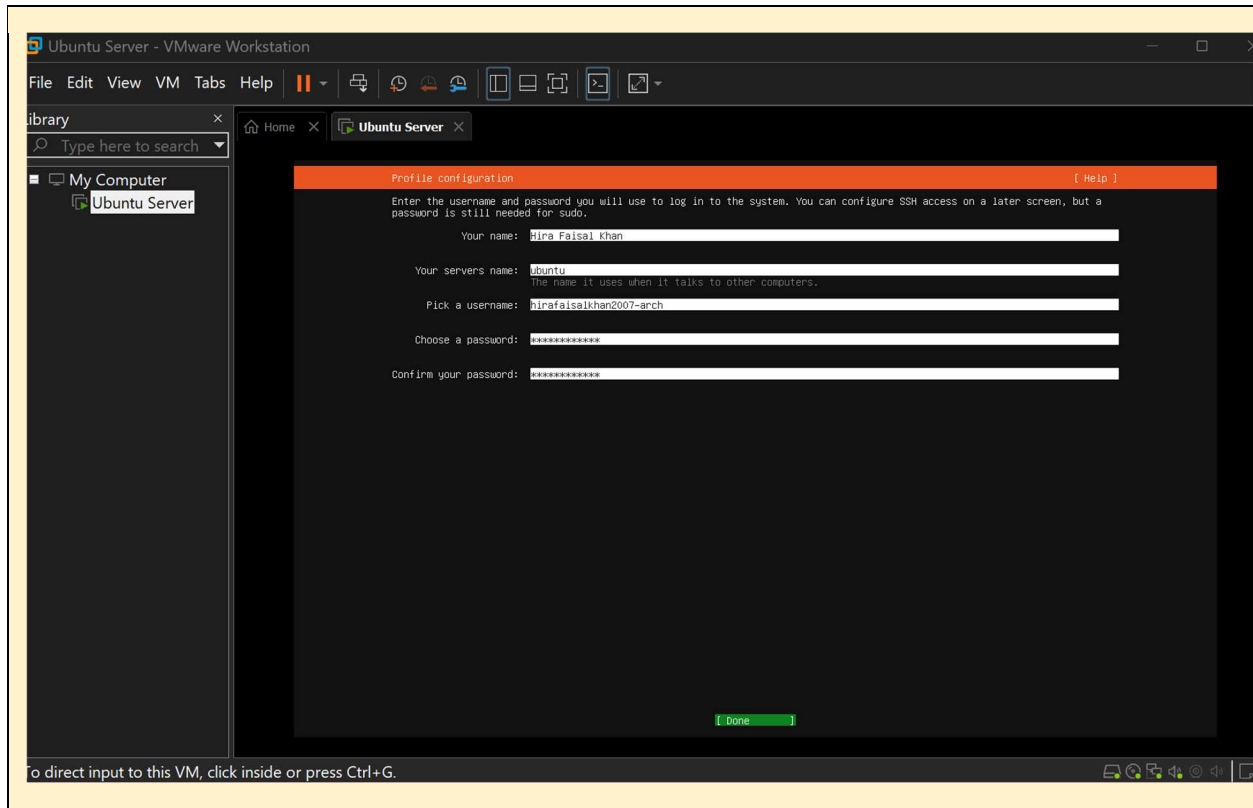
Caption: This screenshot shows 'ubuntu' entered in the 'Your server's name' field.

File: ubuntu_username.png



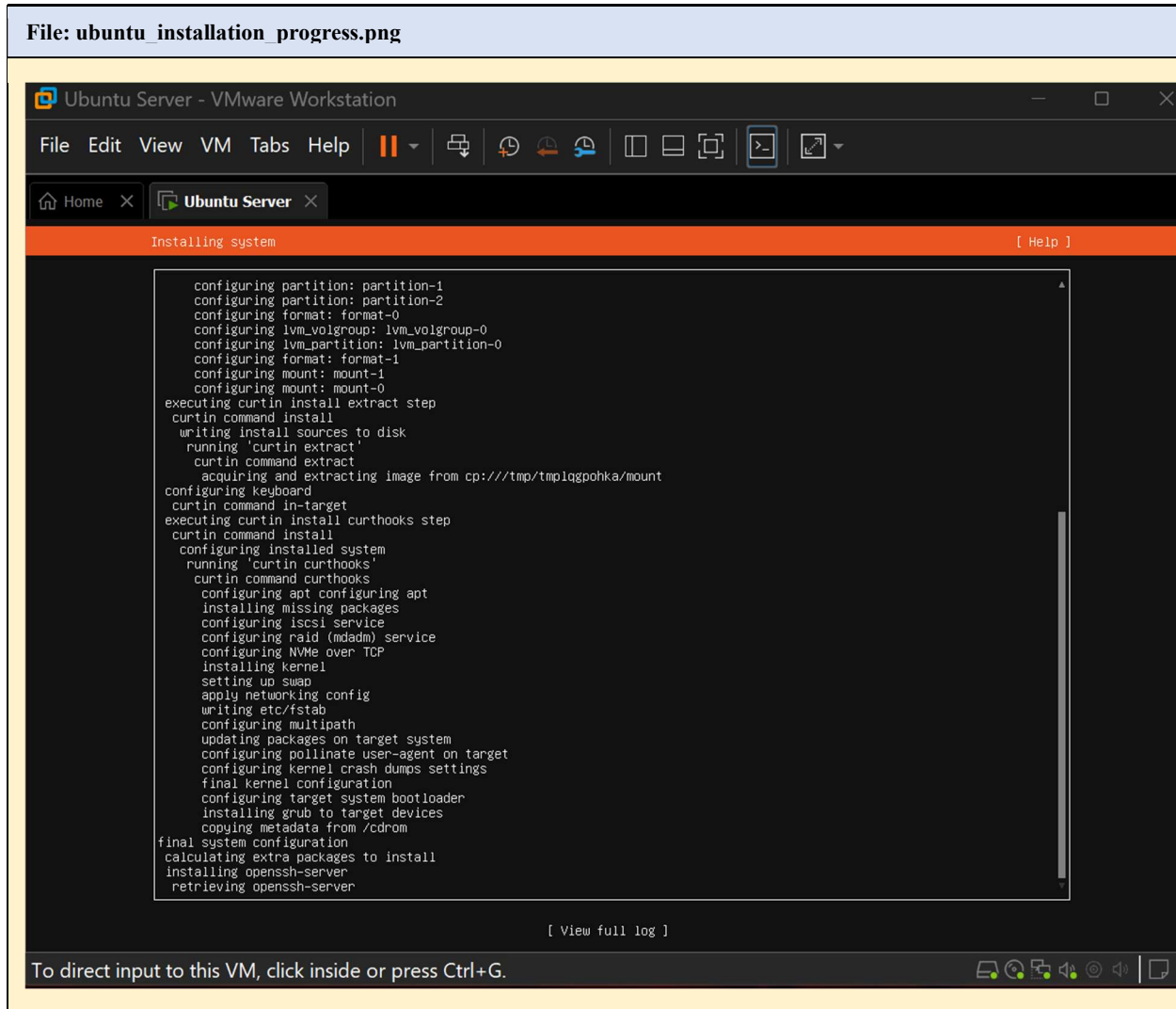
Caption: This screenshot shows my exact GitHub username entered in the 'Pick a username' field.

File: ubuntu_profile_setup.png



Caption: This screenshot shows the completed Ubuntu profile setup page with all fields reviewed.

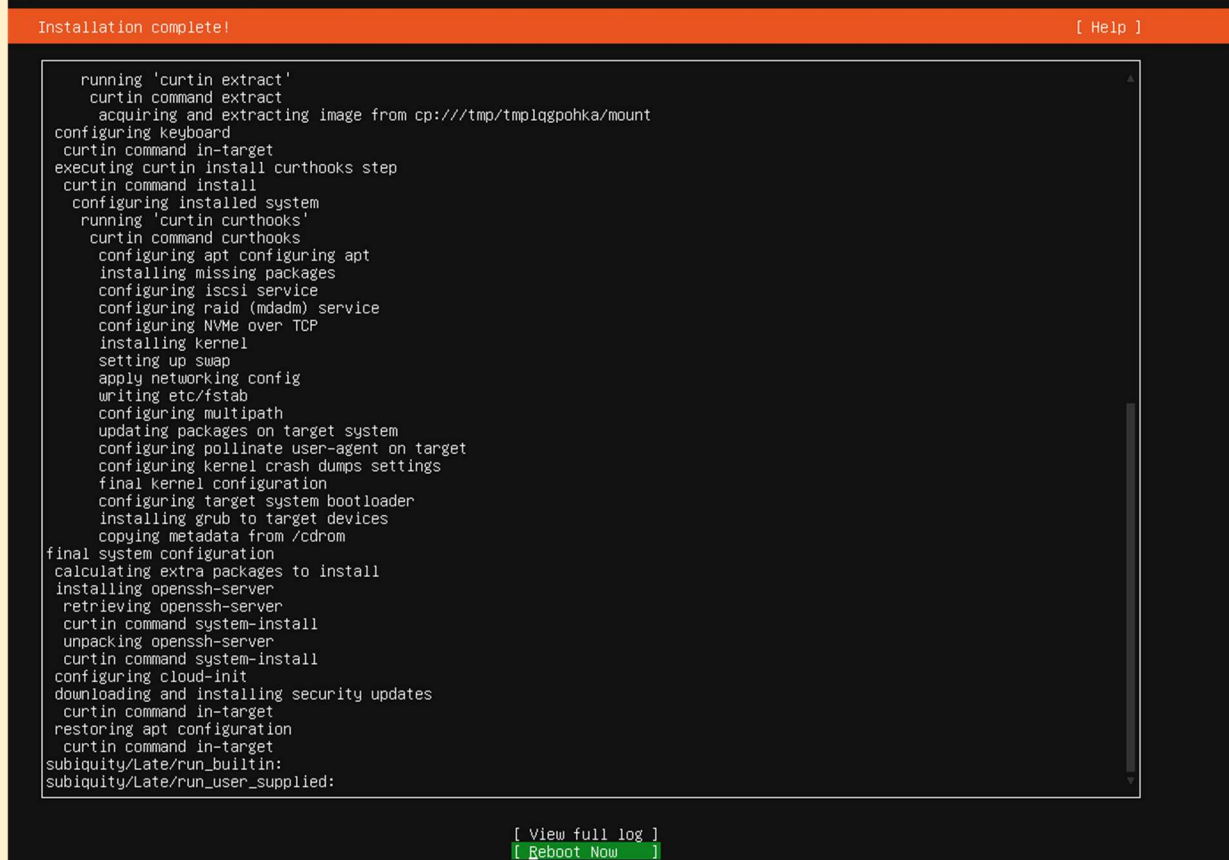
Task 14: Complete the Software Installation



Caption: This screenshot shows the Ubuntu Server software installation in progress.

Task 15: Reboot and Verify the Ubuntu Identity

File: ubuntu_installation_complete.png



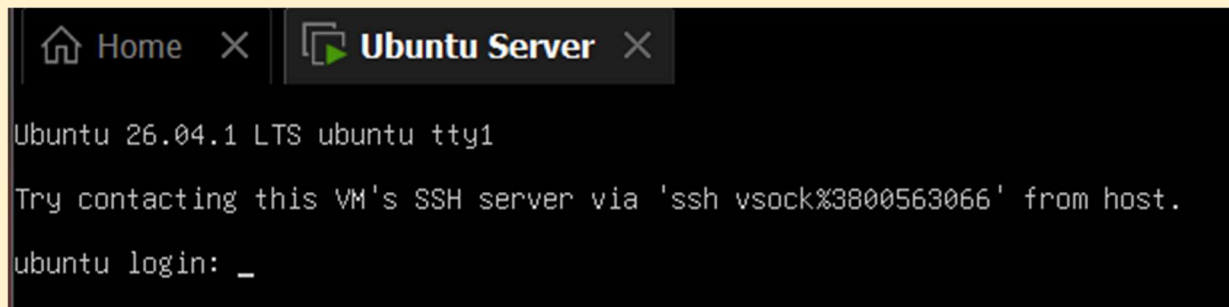
```
Installation complete! [ Help ]

running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmp1qgpohka/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:

[ View full log ]
[ Reboot Now ]
```

Caption: This screenshot shows the installation-complete screen before selecting Reboot Now.

File: ubuntu_login_screen.png



```
Home X Ubuntu Server X

Ubuntu 26.04.1 LTS ubuntu tty1

Try contacting this VM's SSH server via 'ssh vsock%3800563066' from host.

ubuntu login: _
```

Caption: This screenshot shows the Ubuntu Server login screen after rebooting.

File: ubuntu_terminal_login.png

```
hirafaisalkhan2007-arch x + v
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\User> ssh hirafaisalkhan2007-arch@192.168.132.129
The authenticity of host '192.168.132.129 (192.168.132.129)' can't be established.
ED25519 key fingerprint is SHA256:Jg9Ig/DKhuONNlsEdTXZrd3beZZd/6cWFgdzxbXwYqk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.132.129' (ED25519) to the list of known hosts.
hirafaisalkhan2007-arch@192.168.132.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 11:54:16 PM UTC 2026

System load:  0.16           Processes:            219
Usage of /:   50.6% of 9.75GB Users logged in:        0
Memory usage: 22%           IPv4 address for ens33: 192.168.132.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

hirafaisalkhan2007-arch@ubuntu:~$ |
```

Caption: This screenshot shows a successful terminal login using the configured Ubuntu username and password.

```
File: ubuntu_identity_verified.png

hirafaisalkhan2007-arch@ubuntu:~$
hirafaisalkhan2007-arch@ubuntu:~$
hirafaisalkhan2007-arch@ubuntu:~$ _
```

Caption: This screenshot shows the terminal prompt matching the required format <github-username>@ubuntu:~\$.

Task 16: Find the Ubuntu Server IP Address

File: ubuntu_ip_addr_command.png

```
hirafaisalkhan2007-arch@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:87:fd:7a brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnamename enx000c2987fd7a
    inet 192.168.132.129/24 metric 100 brd 192.168.132.255 scope global dynamic ens33
        valid_lft 1721sec preferred_lft 1721sec
    inet6 fe80::20c:29ff:fe87:fd7a/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
hirafaisalkhan2007-arch@ubuntu:~$
```

Caption: This screenshot shows the 'ip addr' command run in the terminal along with its output.

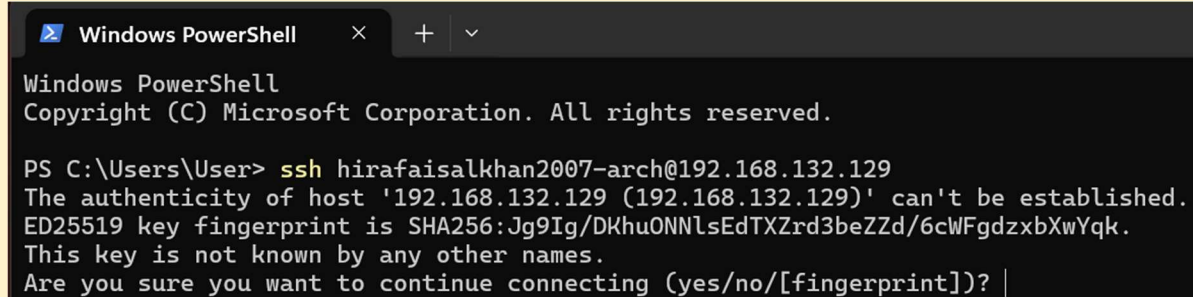
File: ubuntu_ip_address.png

```
    inet 192.168.132.129/24 metric 100 brd 192.168.132.255 scope global dynamic ens33
        valid_lft 1721sec preferred_lft 1721sec
    inet6 fe80::20c:29ff:fe87:fd7a/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
hirafaisalkhan2007-arch@ubuntu:~$ _
```

Caption: This screenshot clearly shows the correct IPv4 address (not the loopback address) with the terminal prompt visible.

Task 17: Connect to Ubuntu Server from Windows

File: windows_ssh_command.png

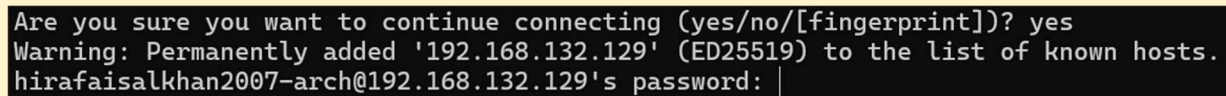


```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\User> ssh hirafaisalkhan2007-arch@192.168.132.129
The authenticity of host '192.168.132.129 (192.168.132.129)' can't be established.
ED25519 key fingerprint is SHA256:Jg9Ig/DKhuONNlsEdTXZrd3beZZd/6cWFgdzxbXwYqk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

Caption: This screenshot shows the SSH command being run from Windows Command Prompt/PowerShell to connect to the Ubuntu server.

File: windows_ssh_fingerprint.png



```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.132.129' (ED25519) to the list of known hosts.
hirafaisalkhan2007-arch@192.168.132.129's password: |
```

Caption: This screenshot shows the server fingerprint prompt accepted by typing 'yes'.

File: windows_ssh_login.png

```
hirafaisalkhan2007-arch@192.168.132.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 11:54:16 PM UTC 2026

System load:  0.16           Processes:            219
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Memory usage: 22%           IPv4 address for ens33: 192.168.132.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

hirafaisalkhan2007-arch@ubuntu:~$ |
```

Caption: This screenshot shows a successful SSH session established after entering the Ubuntu password.

File: windows_ssh_identity.png

```
hirafaisalkhan2007-arch@ubuntu:~$ |
```

Caption: This screenshot clearly shows the remote SSH prompt displaying <github-username>@ubuntu, confirming the connection.